INFORMATION TO KEEP WITH YOU

My complete treatment handbook

AC · every 3 weeks

Understand your treatment. Know what to expect. Keep your plan to hand.

Displayed calendar
21 days
See the active treatment phase

Treatment stages
1
Explained in the treatment reference

Regimen source
NCCP 00252
Version 7

> Doxorubicin and cyclophosphamide on Day 1 every 21 days for four cycles. Any subsequent taxane follows a separately agreed regimen.

Treatment stages

01 **Phase 1 · AC**
21-day interval

Your team confirms the active stage and dates. The full treatment guide explains each step.

Inside your handbook

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My name _____

Hospital / oncology team _____

Use the urgent instructions and contact numbers in your treatment guide. Your team must replace example numbers before patient use. Your individual treatment plan takes precedence.

CHAPTER 01

Understanding your treatment

IN THIS CHAPTER

Why this treatment is offered, what its medicines do, and how the agreed plan fits your care.

AC · every 3 weeks
National Cancer Control Programme (NCCP) 00252

When to get help

Emergency: call 112 / 999

Severe breathing difficulty, collapse, seizure or severe chest pain: call emergency services. Do not wait for an online answer.

Contact your oncology service immediately

Temperature 37.5°C or higher, 36°C or lower, shivering or suddenly feeling unwell — even with a normal temperature. Report new breathing symptoms, severe diarrhoea, inability to drink, painful swollen limbs, unusual bleeding, or worsening numbness. Tell staff immediately about infusion-site pain or an allergic reaction.

Follow the urgent contact instructions from your treating hospital. If you cannot reach the team, seek urgent medical assessment. Outside clinic hours, use your instructed out-of-hours pathway or the emergency department.

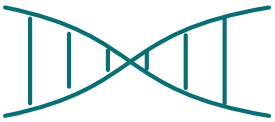
This page does not diagnose symptoms or decide whether it is safe to stay at home. You do not need to know which medicine caused a symptom before calling.

How the medicines work

Treatment after surgery to reduce the chance of breast cancer returning.

ANTHRACYCLINE CHEMOTHERAPY

Doxorubicin



Interferes with deoxyribonucleic acid (the cell's genetic material) (DNA) handling

Doxorubicin interferes with enzymes and DNA processes needed for cell division. It can also affect heart muscle, so the team considers previous exposure and cardiac risk.

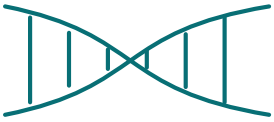
DNA handling disrupted → Cell damage → Division reduced

What to watch for

Report pain or burning at the infusion site immediately. New breathlessness or chest symptoms need prompt review. Red urine for a short time can be a drug-colour effect; pain, clots or persistent bleeding still need advice.

ALKYLATING CHEMOTHERAPY

Cyclophosphamide



Damages DNA needed for growth

Cyclophosphamide is activated in the body and damages DNA. This helps stop dividing cancer cells but can also affect blood-cell production and reproductive tissues.

Drug activated → DNA damage → Cell division affected

What to watch for

Use the prescribed anti-sickness and hydration plan. Report fever, painful urination or blood in the urine.

The drawings are simplified explanations of drug action, not anatomical pictures or a prediction of response.

Why this treatment?

AC may be selected after surgery to reduce the chance of cancer returning. Cancer size, nodes, grade, hormone receptors, HER2 status, health and preferences influence the choice. AC alone is a four-cycle regimen; a later taxane is a separate planned regimen, not an automatic extra phase.

Your treatment setting

Adjuvant breast cancer

The sequence

Doxorubicin and cyclophosphamide on Day 1 every 21 days for four cycles. Any subsequent taxane follows a separately agreed regimen.

Alternatives and your priorities

Ask what alternatives fit your cancer, what the treatment aims to achieve, and how its practical demands fit your life. The team can discuss personal benefits and risks.

Why this treatment may be chosen

Show trial results and what the numbers mean

Adjuvant breast cancer

- Pembrolizumab + chemotherapy
- Placebo + chemotherapy

Your team can discuss expected benefit and alternatives in the context of your own cancer and priorities.

CHAPTER 02

Your schedule, tests and appointments

IN THIS CHAPTER

Your treatment stages, checks and practical preparations, from before treatment through follow-up.

AC · every 3 weeks
NCCP 00252

The phases and how they connect

AC · 21-day reference cycle

21-day view



This shows treatment timing, not guaranteed symptom-free days. Your team confirms the active phase and dates.

21-DAY INTERVAL

Phase 1 · AC

4 cycles

Four AC cycles 21 days apart. Growth-factor use follows the regimen and individual risk plan.

The treatment phase determines the medicines, supportive prescription and tests. A phase change does not reset previous drug exposure or cardiac monitoring.

BASELINE → TREATMENT → FOLLOW-UP

When checks happen

Relative milestones, not equal time intervals. The clinical team owns booking, result review and action on overdue tests. This is an educational map, not an automated appointment or result tracker.

The team arranges and reviews tests. This guide shows expected timing; it does not record completed appointments. NCCP 00252 v7 · official regimen · ESMO 2024 early breast cancer guideline · treatment and follow-up · ESC 2022 cardio-oncology guideline · cardiac surveillance and risk · NICE NG101 · bone health and follow-up · ESMO metastatic breast cancer guideline · 2026

THE WHOLE TREATMENT JOURNEY

AC · every 3 weeks: treatment and checks

Adjuvant breast cancer

Read from left to right. Columns show stages, not equal lengths of time. This is a general plan: no dates or personal information are needed. Your team confirms which steps apply and arranges appointments.

Treatment, tests and reviews across the course

Care / check	Before starting	Early cycles	Later cycles / transition	Next treatment / completion	Follow-up
Cancer medicines NCCP 00252	Agree aim, indication and medicines	AC · every 3 weeks starts	Review response, toxicity and exposure	Complete planned course / next decision	Ongoing review / subsequent plan
Surgery EARLY	Adjuvant: surgery already completed	No automatic operation during cycles	Review response / surgical plan	Surgery after neoadjuvant therapy, if applicable	Pathology and recovery guide next steps

Care / check	Before starting	Early cycles	Later cycles / transition	Next treatment / completion	Follow-up
Radiotherapy EARLY	Assess indication	Usually after cytotoxic chemotherapy	Arrange sequential timing	Radiation team confirms start	Review effects / follow-up
Blood tests NCCP 00252	full blood count (FBC), kidney and liver profile	Review before chemotherapy cycles	Check again at any phase change	Follow active medicines and abnormalities	Follow unresolved abnormalities
Echo / heart NCCP 00252 · ESC	Baseline echo (ESC); assess risk	High cardiac risk: repeat after 2 anthracycline cycles	If continuing: repeat after 4 cycles at high risk	Review cardiac symptoms and total lifetime exposure	Echo within 12 months; high risk within 3 months
electrocardiogram (ECG) NCCP 00252	Baseline ECG in the NCCP anthracycline source; repeat for symptoms or clinical risk. An ECG does not replace an echo.	Symptoms / risk-based repeat	Review if cardiac concern	ECG and echo answer different questions	Clinical indication
Cancer imaging NCCP 00252	Record baseline extent as appropriate	Clinical response / symptoms review	If before surgery: plan response review	Arrange selected surgery / next treatment	Planned breast imaging; no routine computed tomography (CT)/positron emission tomography (PET)

Care / check	Before starting	Early cycles	Later cycles / transition	Next treatment / completion	Follow-up
Markers / bone health NCCP 00252	Tests selected for active medicines	No routine tumour-marker clearance	No routine blood tumour markers when well	dual-energy X-ray absorptiometry (bone density) (DEXA) if endocrine/ bone-risk indication	Follow the endocrine and bone-health plan
Endocrine / bone BONE · EARLY	Confirm hormone receptors	Plan endocrine therapy when appropriate	Separate from this chemotherapy prescription	Start/continue per multidisciplinary team (MDT); DEXA when indicated	Risk-based bone review

What each check is for

1 Before first dose

Baseline

Baseline FBC, kidney and liver tests; repeat before each chemotherapy cycle.

Ask / act: Confirm tests and consent before prescribing.

2 Heart-monitoring plan

Baseline → cycles → after treatment

ESC supplementary guidance: baseline echo before anthracycline treatment. High/very-high cardiac risk: echo every two anthracycline cycles (for example before cycles 3 and 5 when continuing), and within 3 months after completion. All adults: echo within 12 months after completion. Other repeat tests depend on risk, cumulative exposure and symptoms.

Ask / act: Name the team responsible for booking and reviewing results.

3

Treatment-phase review

Before each transition

Doxorubicin and cyclophosphamide on Day 1 every 21 days for four cycles. Any subsequent taxane follows a separately agreed regimen.

Ask / **act:** Confirm active medicines, support script and the next test checkpoint.

4

Response and follow-up

Planned review points

For early breast cancer, breast imaging and clinical review follow the agreed pathway. Routine CT/PET scans and blood tumour markers are not used to look for distant recurrence when there are no symptoms or clinical concerns.

Ask / **act:** Review symptoms and imaging together; do not change treatment on tumour markers alone.

At the next review, ask

- Which treatment phase am I in, and what happens next?
- Which test is expected next, who arranges it, and who explains the result?
- Who should I contact if an expected appointment has not arrived?

This map does not show whether a test has been ordered, booked or completed. Do not wait for a scheduled test if you develop new symptoms.

Your practical treatment plan

STEP 1

Before the first visit

Blood tests, relevant cardiac assessment, medicines review, fertility discussion and the consent conversation come first.

STEP 2

Plan venous access

A cannula, peripherally inserted central catheter (PICC) or implanted port may be used. If a port is chosen, a trained team inserts it in a separate procedure with local anaesthetic and sometimes sedation; the team explains fasting, blood-thinner and wound instructions.

STEP 3

Infusion day

Staff check the line, give prescribed premedicines and treatment, and observe for reactions. Tell them immediately about burning, pain, rash or breathlessness.

STEP 4

Home and next review

This regimen does not routinely need a take-home infusion pump. Staff disconnect the infusion; a port is de-accessed when appropriate. Keep line-care and next-visit instructions.

Port insertion and removal

The line team explains placement, dressing care and when normal activities can resume. A port is accessed with a special needle during treatment. Removal of the implanted device is a separate planned procedure when no longer needed; do not remove or adjust it yourself.

PICC care and urgent line symptoms

A PICC usually needs regular dressing and flushing appointments even between treatments. Fever, redness, swelling, pain, leakage or damage requires prompt advice. Your hospital supplies the exact care instructions.

Visit time and transport

First visits can take longer because of loading doses, premedicines and observation. Ask the unit about expected time, transport and any escort needed after sedation or line insertion.

Phase-specific visit pattern

Infusion and observation times differ by drug and first versus later doses; the unit confirms the full visit duration, including blood review and preparation.

Checks across treatment

Your team arranges, reviews and acts on tests; an overdue result needs follow-up.

Blood tests

Baseline FBC, kidney and liver tests; repeat before each chemotherapy cycle.

Heart monitoring

ESC supplementary guidance: baseline echo before anthracycline treatment. High/very-high cardiac risk: echo every two anthracycline cycles (for example before cycles 3 and 5 when continuing), and within 3 months after completion. All adults: echo within 12 months after completion. Other repeat tests depend on risk, cumulative exposure and symptoms.

ECG

Baseline ECG in the NCCP anthracycline source; repeat for symptoms or clinical risk. An ECG does not replace an echo.

Cancer scans

For early breast cancer, breast imaging and clinical review follow the agreed pathway. Routine CT/PET scans and blood tumour markers are not used to look for distant recurrence when there are no symptoms or clinical concerns.

Surgery, radiotherapy and later treatment

Surgery → this adjuvant chemotherapy → radiotherapy if indicated. The surgical team decides lumpectomy versus mastectomy; the radiation plan depends on surgery, pathology and individual risk.

Hormone-receptor-positive disease

Endocrine treatment is considered when appropriate for the cancer biology and setting. It has its own schedule and monitoring; it is not automatically part of this chemotherapy prescription.

Bone health

DEXA is linked to an aromatase inhibitor, ovarian suppression, treatment-induced menopause or other bone risk—not automatically to every chemotherapy course. The endocrine team confirms baseline testing and risk-based repeat scans.

Markers and follow-up

After local treatment, the team plans mammography of remaining breast tissue. A mastectomy side does not need routine mammography. Report new symptoms rather than waiting for a scan.

Genomic tests

Where appropriate, the MDT may discuss a validated genomic test to inform the chemotherapy decision. This regimen card does not calculate an Oncotype result or select treatment from a score.

CHAPTER 03

Symptoms and supportive care

IN THIS CHAPTER

Recognising changes, knowing when to call, and using your prescribed supportive-care plan.

AC · every 3 weeks
NCCP 00252

AC · 21-day reference cycle

Day 1 → Day 21: a general pattern

This is a visual orientation, not a prediction of your personal cycle. It shows when some effects often emerge and which effects are **not** tied to a particular day.

AC · 21-day reference cycle

Treatment or check	Timing in this view
Treatment · Day 1	 **Day 1** intravenous (IV)
Administration reaction	 **Days 1–21** Report during or after any visit
Nausea / appetite	 **Days 1–5** May occur early or later
Blood counts / infection	 **Days 1–21** Risk varies; fever needs prompt advice
Fatigue / mouth changes	 **Days 1–21** Variable timing and severity

No dates to enter: The chart explains the pattern. Your team books actual appointments and reviews results.

Phase changes: Do not change the schedule using this educational map.

Treatment delays: A delayed dose does not automatically delay an echo or other test that is already due.

Why these bands stay broad: These broad symptom bands are educational possibilities, not measured risk curves. Infusion days are the planned schedule; delays require a new clinical plan. Band length does not show likelihood or severity, and the chart cannot tell you whether it is safe to receive treatment. NCCP 00252 v7 · official regimen ↗

There are no guaranteed "safe days". Fever, shivers, sudden illness, severe breathing changes or chest pain need prompt assessment on any day.

Between-treatment notes

Day 1–21: Your current reference cycle

Doxorubicin and cyclophosphamide on Day 1 every 21 days for four cycles. Any subsequent taxane follows a separately agreed regimen. Follow the team's appointment and medicine instructions; contact them about missed doses or new symptoms.

Support around the medicines

1 

The correct phase prescription

AC uses the high-risk antiemetic plan.

2

Premedicines and growth factors

Confirm antiemetic and growth-factor prescriptions before leaving the unit.

3

Tests and appointments

The team should name the booking owner, due window and person reviewing each required test. Tell them if an expected appointment has not arrived.

4

After chemotherapy

Discuss recovery, fertility, rehabilitation and the next treatment or surveillance step.

Questions worth asking

When are heart checks due?

Baseline cardiac assessment before AC. Repeat imaging according to symptoms, anthracycline exposure and cardiac risk; no universal 12-week echo schedule is assigned to AC alone. ESC guidance supplements the source: baseline echo before anthracyclines; high / very-high cardiac risk may need echo every two cycles and within 3 months after treatment. Test timing follows the treatment, clinical findings and local pathway.

Where do surgery and radiotherapy fit?

In adjuvant care, radiotherapy usually follows chemotherapy when indicated; endocrine therapy follows its own receptor-specific plan.

Do I need routine CT scans or tumour markers?

Early breast follow-up uses planned breast imaging. Routine CT / PET, blood tumour markers or other distant-recurrence screening are not recommended without symptoms or a clinical indication. After local treatment: annual mammography of remaining breast tissue; additional imaging when indicated.

Does a port mean a pump at home?

No. This regimen has no routine take-home infusion pump. A port is an access device; the team removes the needle after the visit when appropriate and arranges device removal separately.

Who arranges the scans and tests?

Your treating service should book the planned checks, review the results and explain any change. If an expected appointment is missing, contact the team; this map does not book tests or confirm that they have happened.

> **Try explaining the plan back:** Can you explain which medicines you are receiving now? What symptoms mean you should call urgently? Which blood tests, scans or heart checks come next, and who arranges them?

Doxorubicin

- Low counts, infection and mouth ulcers
- Nausea, hair loss and fatigue
- Heart damage, sometimes delayed
- Severe tissue injury if the infusion leaks
- Rare later blood cancers

Cyclophosphamide

- Low counts and infection
- Nausea, vomiting and hair loss
- Bladder irritation / bleeding
- Fertility effects and menopause

Supportive prescription draft for review. This is a regimen template. It is not an issued patient prescription; individual review and completion are required.

AC · 21-day schedule

Active medicines: Doxorubicin + Cyclophosphamide

Emetogenic risk: High · AC

Medicine	Dose / route	When
Aprepitant	125 mg orally	Day 1
Ondansetron	16 mg orally	Day 1 Individual dose / QT / hepatic review
Dexamethasone	12 mg orally	Day 1 No routine Days 2–4 dexamethasone in this NCCP AC antiemetic row

Medicine	Dose / route	When
Olanzapine	5 mg orally	Days 1–4 Review sedation, falls, interactions and glucose
Aprepitant	80 mg orally daily	Days 2–3
Metoclopramide	10 mg orally up to three times daily	Subsequent days: duration and use to be confirmed Caution with olanzapine: additive extrapyramidal effects; adjust / avoid for individual contraindications

Additional phase-specific support

- Assess granulocyte colony-stimulating factor (white-cell growth support) (G-CSF) according to source guidance and individual risk.
- Use this script for the prescribed AC cycles; review continuing symptoms and rescue medicines after completion.

Before signing: verify allergies, current medicines, interactions, renal / hepatic function, QT risk, sedation, prior response, total steroid exposure and the rescue-medicine duration. Do not duplicate medicines from another phase.

Individual changes / rescue duration: _____
Prescriber / date: _____ Pharmacy review: _____

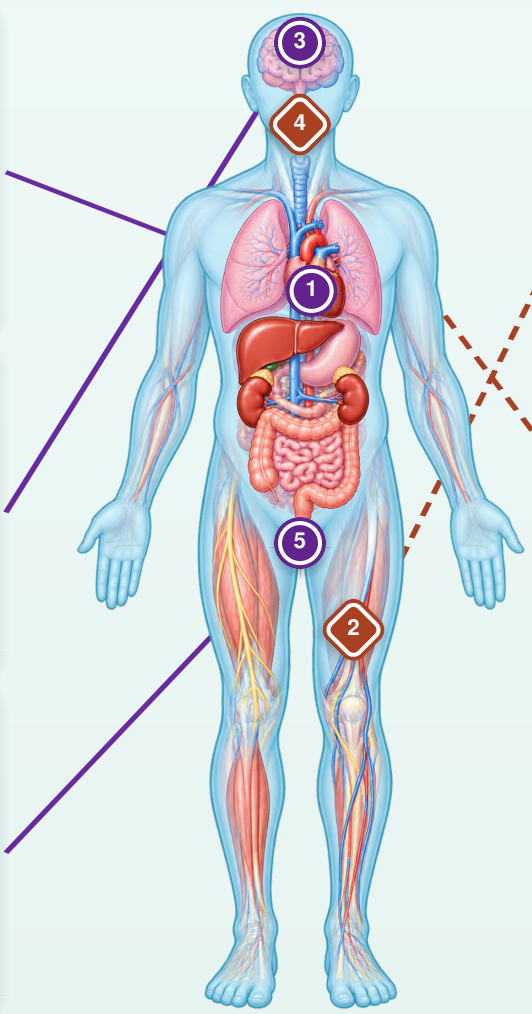
AC · every 3 weeks

Doxorubicin and cyclophosphamide on Day 1 every 21 days for four cycles. Any subsequent taxane follows a separately agreed regimen.

Source reconciled; independent clinical, pharmacy and patient/PPI review pending. Selected effects; not a complete medicine leaflet.

A picture of side effects

Numbered symptom boxes connect to the anatomical illustration. The medicine responsible may be uncertain; report the symptom.



1 Heart

Doxorubicin

Baseline assessment; report chest pain, breathlessness or swelling.

2 Blood / infection

Chemotherapy phase

Fever, shivers or suddenly unwell: call immediately.

3 Hair / scalp

Chemotherapy phase

Hair loss is common. Ask about scalp cooling before treatment where suitable.

4 Mouth

Chemotherapy phase

Painful ulcers or difficulty drinking need advice.

5 Bladder

Cyclophosphamide

Report blood in urine or painful urination.

Selected effects, not every possible effect or a prediction. Blood-count, skin and line markers are schematic. Number and text carry meaning; colour is decorative.

Selected medicine-specific effects are highlighted. The map is not a complete list; persistent or new symptoms need review even after a medicine stops.

> Fever, shivers or suddenly unwell: contact oncology immediately. Severe chest pain, breathing difficulty, collapse or seizure: 112 / 999.

CHAPTER 04

Sources and further information

IN THIS CHAPTER

The references behind this information and addresses you can use to find further help.

Use this information alongside the plan agreed with your treatment team. Your team confirms your medicines, doses and appointments.

1. **Regimen selection and source references**
   https://healthservice.hse.ie/documents/6356/252_DOXO_cycloPHOS_AC.pdf
2. **ESMO 2024 early breast cancer guideline · treatment and follow-up**
   https://www.annalsofoncology.org/article/S0923-7534(23)05104-9/fulltext
3. **ESC 2022 cardio-oncology guideline · cardiac surveillance and risk**
   https://academic.oup.com/eurheartj/article/43/41/4229/6673995
4. **NICE NG101 · bone health and follow-up**
   https://www.nice.org.uk/guidance/ng101/chapter/Recommendations
5. **ESMO metastatic breast cancer guideline · 2026**
   https://www.annalsofoncology.org/article/S0923-7534(26)00902-6/fulltext
6. **eviQ · implanted port patient information**
   https://www.eviq.org.au/p/507
7. **eviQ · peripherally inserted central catheter (PICC) patient information**
   https://www.eviq.org.au/p/543
8. **HSE · chemotherapy side effects and urgent advice**
   https://www2.hse.ie/conditions/chemotherapy/side-effects/
9. **NOGG · section 7, aromatase inhibitors and fracture risk**
   https://www.nogg.org.uk/full-guideline/section-7-strategies-management-osteoporosis-and-fracture-risk
10. **NCCP NCIS antiemetics v8 (2025), Table 1**
    https://healthservice.hse.ie/documents/7180/Supportive_Care_Antiemetics_for_inclusion_NCIS_Medical_Oncology.pdf
11. **ASCO · biomarkers in metastatic breast cancer**
    https://ascopubs.org/doi/10.1200/JCO.2015.61.1459